Icarus and Daedalus: Non-Gaussian Size-Dependent Productivity and Aggregate Risk

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A common benchmark for productivity

Linear AR(1) log productivity with Gaussian, size-independent innovations

Hugo Hopenhayn and Richard Rogerson (1993)
Job Turnover and Policy Evaluation: A General Equilibrium Analysis

$$\log(s_t) = a + \rho \log(s_{t-1}) + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2), a \geq 0, 0 \leq \rho < 1,$$

Virgiliu Midrigan and Daniel Yi Xu (2014)
Finance and Misallocation: Evidence from Plant-Level Data

We assume that the process for transitory productivity evolves according to an AR(1) process with Gaussian disturbances

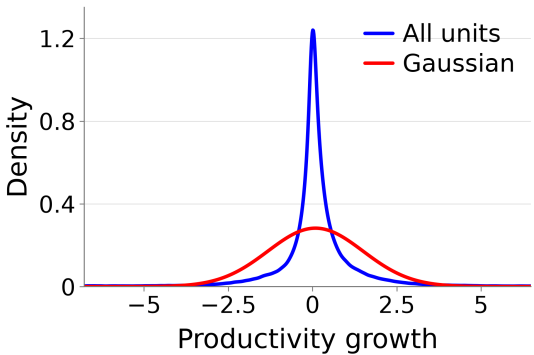
Thomas Winberry (2021)
Lumpy Investment, Business Cycles, and Stimulus Policy

$$\log \varepsilon_{jt+1} = \rho_\varepsilon \log \varepsilon_{jt} + \omega_{t+1}^{\varepsilon}, \quad \text{where} \quad \omega_{t+1}^{\varepsilon} \sim N\left(0, \sigma_\varepsilon^2\right).$$

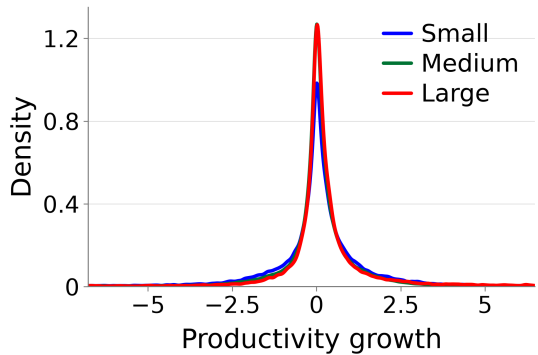
Normality: Is the conditional productivity distribution Gaussian?
Size dependence: Does the transition law vary with producer size?

Question 1: Micro Law of Motion for Productivity Growth

A sharply peaked distribution



Shape varies with size



Distribution: How does productivity risk vary with producer size?

Dynamics: What follows a gain or loss—persistence, reversal, or recovery?

What We Measure, Estimate, and Find

Data and measurement

Restricted-use Indian manufacturing (ASI) plant–product data, 2010–11 to 2022–23.
Detailed output, input, and price information $\longrightarrow$ **physical-output** productivity growth.

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Microeconomic findings

1. **Shape.** Sharp peak and fat tails.
2. **Daedalian buffering.** Large producers
   - recover more after losses
   - retain more after gains
3. **Nonlinear reversal.**
   - *Icarian liftoff* left-tail large $\downarrow$ buffer
   - *Daedalian corridor* at core $\uparrow$ buffer
   - *Icarian fallout* right-tail small $\downarrow$ buffer
4. **Daedalian range.** Large producers
   - P (left-tail) $\downarrow$ and P (right-tail) $\uparrow$
   - Tail-reach dominates

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Law of motion

Three-regime Markov normal mixture.
Conditioned on history and producer size.

Neutralization	(Core)—flat mean	Low σ
Stabilization	(Shoulders)—recovery	Mid σ
Correction	(Tails)—strong reversal	High σ

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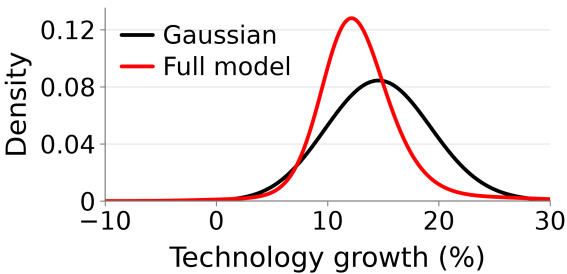
Macroeconomic implications

Cost–Domar aggregation
Maps law of motion into aggregate risk

Question 2: How Do These Dynamics Shape Aggregate Risk?

$$G_{t+1} = \sum_i \underbrace{\widetilde{\lambda}_{i,t}}_{\text{Aggregate exposure}} \underbrace{g_{i,t+1}}_{\text{Productivity dynamics}}$$

Aggregate technology growth



Moments and event probabilities

	Gaussian	Full model
Mean growth	14.69%	12.95%
Standard deviation	4.73 pp	4.25 pp
Growth below 0%	0.055%	0.315%
Growth below 5%	1.81%	1.50%
Growth above 20%	13.05%	4.54%
Far tails: $\lvert z \rvert > 3$	0.29%	1.89%

Whose shocks matter, and what risks do those producers carry?

Lower aggregate volatility coexist with greater contraction and far-tail risk.

Contribution to the Literature

Productivity measurement

Foster, Haltiwanger, and Syverson (2008); Orr (2022); Dhyne, Petrin, Smeets, and Warzynski (2024); McNerney, Savoie, Caravelli, Carvalho, and Farmer (2021); Fadinger, Ghiglino, and Teteryatnikova (2022).

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Granularity and production networks

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Evidence from India's ASI

Hsieh and Klenow (2009, 2014); Boehm and Oberfield (2020, 2023); Cairncross, Morrow, Orr, and Rachapalli (2025).

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Economic Model and Measurement

Measuring physical-output productivity

Production Units and Commodity Aggregation

Competitive CES aggregator

$$\frac{\mathcal{Y}_{g,t}}{\overline{Y}_{g,t}} = \left[\sum_{i \in \mathcal{N}_{g,t}} \sigma_{i,t} \left(\frac{\widetilde{y}_{i,t}}{\overline{\widetilde{y}}_{i,t}} \right)^{\frac{\eta_{g,t}-1}{\eta_{g,t}}} \right]^{\frac{\eta_{g,t}}{\eta_{g,t}-1}}$$

Sales shares $\sigma_{i,t}$; substitution elasticity $\eta_{g,t}$. Bars denote the observed allocation.

4,732 commodities

7-digit commodity codes (2022–23)

e.g. *toothbrush handles* NPCMS 3699016

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Production unit: a plant–product pair

$$\max \sum_{s=0}^{\infty} \frac{\pi_{i,t+s}}{(1+r)^s}$$

subject to demand, technology, and capital accumulation

$$\pi_{i,t} = \underbrace{p_{i,t}(\widetilde{y}_{i,t} + \Delta s_{i,t})}_{\text{Sales + inventory value}} - \underbrace{E^{\text{op}}_{i,t}}_{\text{Operating costs}} - \underbrace{E^{I}_{i,t}}_{\text{Investment}}$$

$$E^{\text{op}}_{i,t} = \sum_f \tau^{\ell}_{if,t} w_{f,t} \ell_{if,t} + \sum_g \tau^{x}_{ig,t} p_{g,t} x_{ig,t}$$

$$E^{I}_{i,t} = \sum_m p_{m,t} \sum_c I_{icm,t}$$

τ: unit-specific input-cost wedges.

Nonparametric production

$$y_{i,t} = F_{i,t}(\ell_{i,t}, \boldsymbol{k}_{i,t}, \boldsymbol{x}_{i,t})$$

Input types and categories, 2022–23

Labor	4
Capital	7
Other domestic inputs	912
Direct imports	3,171
Domestic intermediates	4,035

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Timing

$$\underbrace{y_{i,t}}_{\text{Produced}} = \underbrace{\widetilde{y}_{i,t}}_{\text{Sold}} + \underbrace{\Delta s_{i,t}}_{\text{Inventory change}}$$

$$k_{ic,t+1} = (1 - \delta_{ic,t}) k_{ic,t} + I_{ic,t}$$

$I_{ic,t}$: new capital for the next period.

Cost Wedges and Output Elasticities

Markup–inventory wedge

$$\mu_{i,t} = 1 - \frac{1}{\eta_{g,t}} \frac{y_{i,t}}{\widetilde{y}_{i,t}}$$

$\eta_{g,t}$: substitution within commodity g.

$$\frac{y_{i,t}}{\widetilde{y}_{i,t}} = 1 + \frac{\Delta s_{i,t}}{\widetilde{y}_{i,t}}$$

When $\Delta s_{i,t} = 0$: $\mu_{i,t}$ is the inverse markup.

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Capital user cost

$$R^k_{ic,t} = (1 + r)p^k_{ic,t} - (1 - \delta_{ic,t})p^k_{ic,t+1}$$

Opportunity cost less the resale value of undepreciated capital.

Capital-service expenditure: $R^k_{ic,t} k_{ic,t}$.

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Local output elasticities

Total input cost under CRS:

$$C_{i,t} = \mu_{i,t} p_{i,t} y_{i,t}.$$

FOCs hold inventory changes fixed:

$$\frac{\partial \log y_{i,t}}{\partial \log \ell_{if,t}} = \frac{\tau^{\ell}_{if,t} w_{f,t} \ell_{if,t}}{\mu_{i,t} p_{i,t} y_{i,t}}$$

$$\frac{\partial \log y_{i,t}}{\partial \log k_{ic,t}} = \frac{R^k_{ic,t} k_{ic,t}}{\mu_{i,t} p_{i,t} y_{i,t}}$$

$$\frac{\partial \log y_{i,t}}{\partial \log x_{ig,t}} = \frac{\tau^{x}_{ig,t} p_{g,t} x_{ig,t}}{\mu_{i,t} p_{i,t} y_{i,t}}$$

Effective cost shares give output elasticities.

Physical-Output Productivity Growth

Primary factors, including capital

$$\omega_{i,t}^{L} = 1 - \omega_{i,t}^{X}$$

$$\alpha_{if,t}^{\ell} = \frac{\tau_{if,t}^{\ell} w_{f,t} \ell_{if,t}}{\omega_{i,t}^{L} C_{i,t}}, \qquad \alpha_{ic,t}^{k} = \frac{R_{ic,t}^{k} k_{ic,t}}{\omega_{i,t}^{L} C_{i,t}}$$

Labor, capital, nonmanufactured inputs, and imports.

Manufacturing intermediates

$$\omega_{i,t}^{X} = \frac{\sum_{g} \tau_{ig,t}^{x} p_{g,t} x_{ig,t}}{C_{i,t}}$$

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Production-unit Solow residual

$$g_{i,t+1} = \underbrace{-\omega^L_{i,t}\left(\sum_{f\in\mathcal{F}} \alpha^\ell_{if,t}\Delta_t \frac{\ell_{if}}{y_i} + \sum_{c\in\mathcal{K}} \alpha^k_{ic,t}\Delta_t \frac{k_{ic}}{y_i}\right)}_{\text{Primary factors, including capital}} \underbrace{- \omega^X_{i,t} \sum_{g\in\mathcal{G}} \omega^x_{ig,t}\Delta_t \frac{x_{ig}}{y_i}}_{\text{Manufacturing intermediates}}$$

$\Delta_t z_i \equiv \log z_{i,t+1} - \log z_{i,t}$. All weights are dated t.

Productivity rises when inputs per unit of physical output fall.

Mapping the Data to the Model

Physical quantities, common prices, and unit-specific wedges

Multi-product plants

Keep output at the plant–commodity level. Allocate plant-wide inputs and capital across products using their shares of plant gross sales.

Inventory discrepancies

If $|y_{i,t} - \widetilde{y}_{i,t}|/y_{i,t} > 0.30$, set $\widetilde{y}_{i,t} = y_{i,t}$ and $\Delta s_{i,t} = 0$. Preserve sales and recompute $p_{i,t} = S_{i,t}/\widetilde{y}_{i,t}$.

3.18% of production units affected in 2022–23.

Inverse-markup outliers

Cap $\mu_{i,t}$ at 10 by scaling all costs proportionally.

0.35% of production units affected in 2022–23.

Commodity prices and elasticities

Recover $\eta_{g,t}$ from inverse markups, accounting for inventories. CES aggregation gives $Y_{g,t}$ and $p_{g,t} = S_{g,t}/Y_{g,t}$. Normalize common prices so quantity-weighted mean input wedges equal one.

Input unit-value outliers

Bound τ^x and τ^ℓ in $[0.1, 10]$.

Rescale quantities to preserve expenditure and reference prices.

2022–23 expenditure affected: intermediates 0.051%; primary factors 0.251%.

Capital Measurement: We need a price for capital

Commodity composition of investment

U.S. 1997 capital-flow tables provide weights for seven ASI capital types.

Map them to Indian 7-digit goods using annual sales shares to obtain $\{\alpha^k_{cg,t}\}_g$.

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CES capital prices and quantities

$$p^k_{c,t}(\theta) = \left[\sum_g \alpha^k_{cg,t} p^{1-\theta}_{g,t}\right]^{1/(1-\theta)}$$

$$k_{ic,t}(\theta) = \frac{V^k_{ic,t}}{p^k_{c,t}(\theta)}$$

$V^k_{ic,t}$: closing gross capital value

Benchmark: $\theta = 1$, so $p^k_{c,t} = \prod_g p^{\alpha^k_{cg,t}}_{g,t}$.

Seven capital types

- Land
- Buildings
- Plant & machinery
- Transport equipment
- Computers & software
- Environmental equip.
- Other fixed assets

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Adjacent-period prices

Under stable price assumption

$$p^k_{c,t+1} = p^k_{c,t}$$

Annual capital cost, with $r = 6\%$:

$$R^k_{ic,t} k_{ic,t} = 0.06 V^k_{ic,t} + D_{ic,t},$$

$D_{ic,t}$: reported annual depreciation.

Construct $p^k_{c,t}(\theta)$ separately each year.

Capital prices use a CES composite. Plant production has no imposed parametric form.

Robustness of Productivity Measurement

Main specification **All inputs, $\theta = 1$**

Labor, capital, other domestic inputs, direct imports, and domestic intermediates.

Input coverage: seven alternatives

D.1 Labor

D.2 Labor + capital

D.3 Labor + imports

D.4 Labor + other domestic inputs

D.5 Labor + other domestic inputs + imports

D.6 Labor + domestic intermediates

D.7 All inputs except capital

Capital substitution (3)

D.8–D.10 All inputs, with $\theta \in \{0,\ 0.5,\ 2\}$.

Single-product plants

D.11 All inputs, $\theta = 1$.

One positive-sales product per observed year, 2011–2023.

Main distributional, mean-reversion, and tail patterns recur.

Productivity Facts

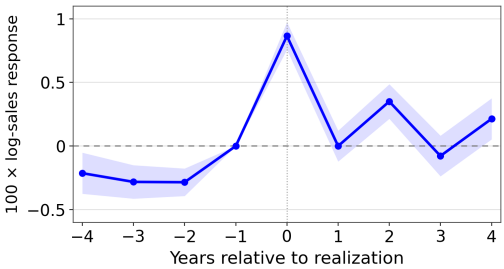
Distributions, dynamics, and firm size

Sales Around Productivity Shocks

Productivity growth

$$\Delta_h \log S_{ipt} = \alpha^h_{pt} + \beta_h g_{ipt} + u^h_{ipt}$$

$$\Delta_h \log S_{ipt} \equiv \log S_{ip,t+h} - \log S_{ip,t-1}$$

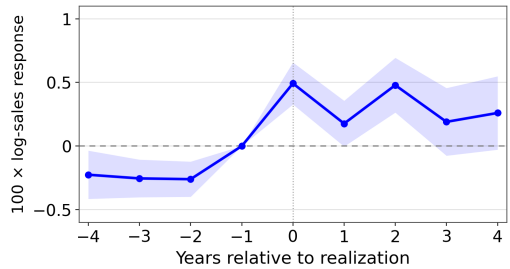


Productivity innovation

$$\varepsilon_{ipt} = g_{ipt} - \widehat{m}_{-k(i)}(g_{ip,t-1}, g_{ip,t-2}).$$

AR(2), five plant folds. Train on four, predict one.

Use ε_{ipt} in the same sales regression.

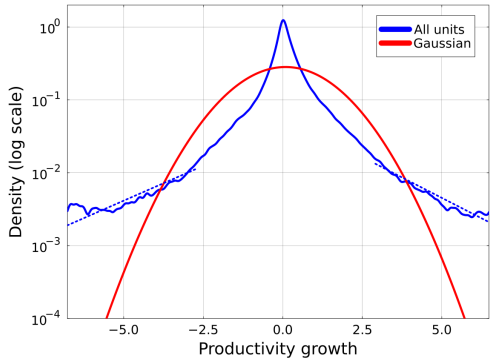


+0.10 log-point realization. Product-by-year fixed effects. Plant-clustered 95% pointwise intervals. Reference: $h = -1$.

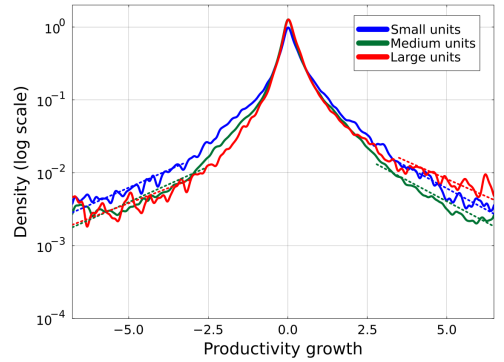
1. **Favorable pre-trends.** Sales are already improving before the productivity realization.
2. **Positive sales responses.** Contemporaneous and persistent.

Non-Gaussian Productivity Growth

Pooled and Gaussian densities



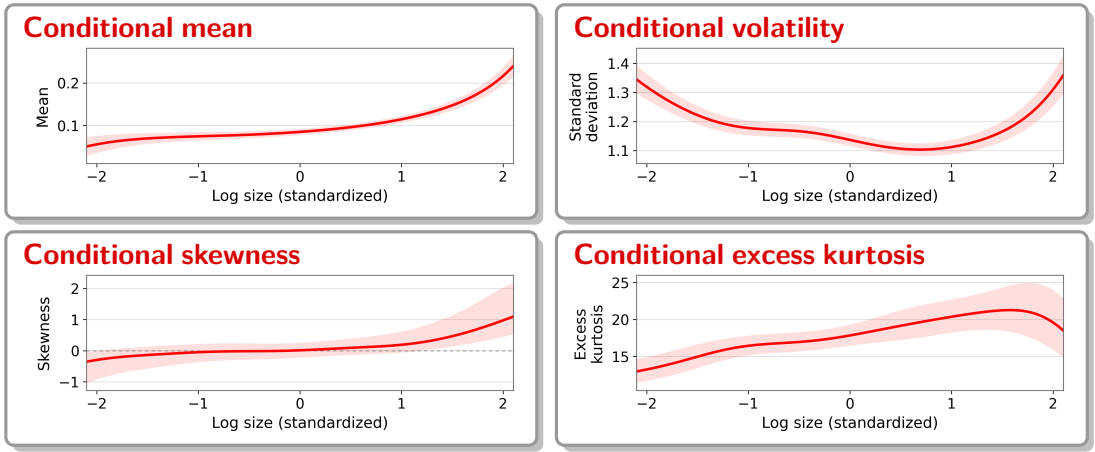
Densities across size groups



Size: standardized beginning-year log sales; cutoffs ± 1.33. Dotted lines: exponential tail fits.

1. **Concentrated core, thin shoulders, fat tails.** Relative to the matched Gaussian.
2. **Tail shape differs across size groups.** Both tails remain prominent at every size.

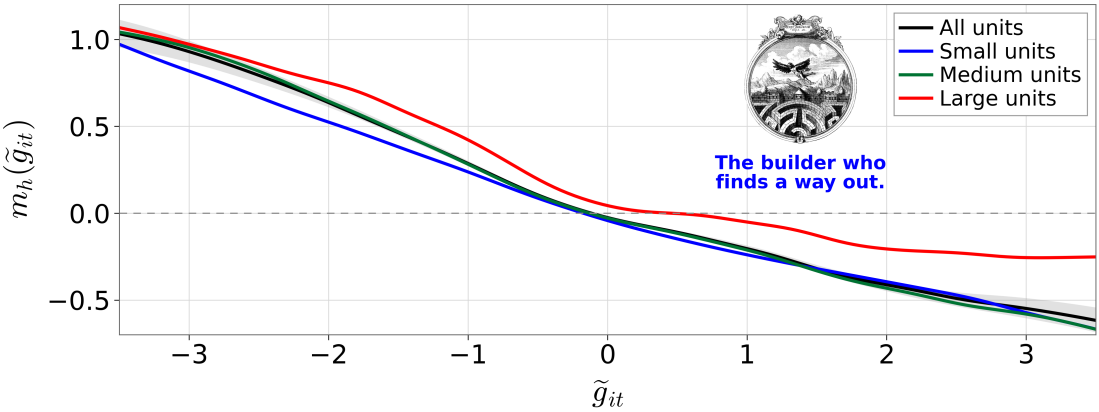
Productivity Growth Across Firm Size



Red: nonparametric estimates. Shading: 95% bands within each panel. Size: standardized log sales.

1. **Mean growth rises; volatility is U-shaped.** Dispersion rises again among the largest units.
2. **Skewness shifts from negative to positive.** Excess kurtosis remains high throughout.

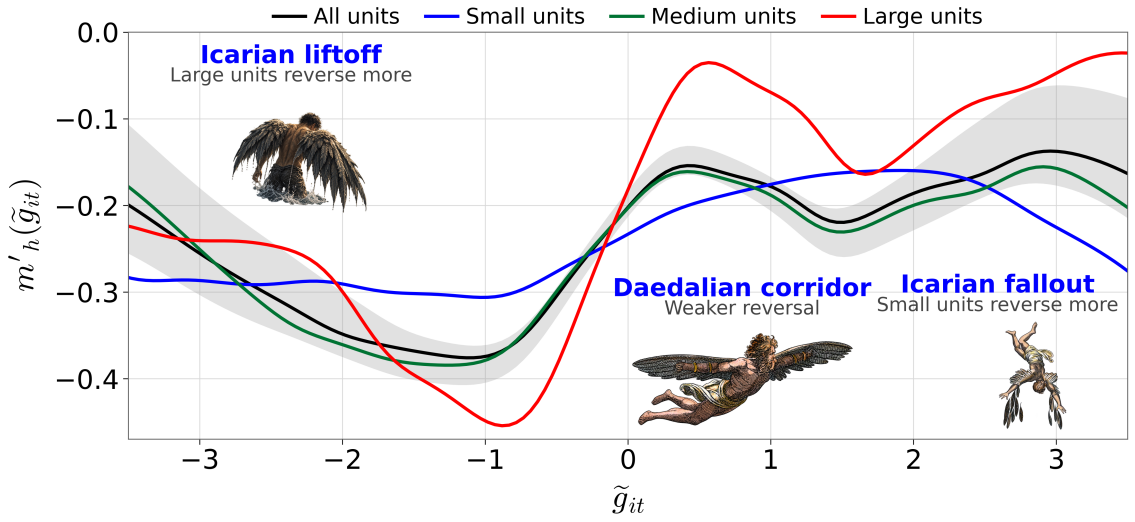
Daedalian Buffering



Tildes denote year-demeaned productivity growth: $\widetilde{g}_{it} = g_{it} - \bar{g}_t$.

1. **Negative shocks: stronger recovery.** Larger firms rebound more strongly.
2. **Positive shocks: greater resilience.** Larger firms retain more of their gains.
3. **Zero crossing farther right.** About $+0.45$ for large units versus -0.17 for small units.

Icarian Liftoff and Fallout



More negative slopes mean stronger marginal reversal in the next year.

Tildes denote year-demeaned productivity growth: $\widetilde{g}_{it} = g_{it} - \bar{g}_t$.

Tail Decay and Firm Size

Beyond each cutoff u: use all observations farther into that tail, with $g = \Delta \log A$.

$Y_L = u - g \ (g < u), \qquad Y_R = g - u \ (g > u), \qquad$ fitted slope $= -\widehat{\zeta}(u) = -\dfrac{1}{\overline{Y}(u)}.$

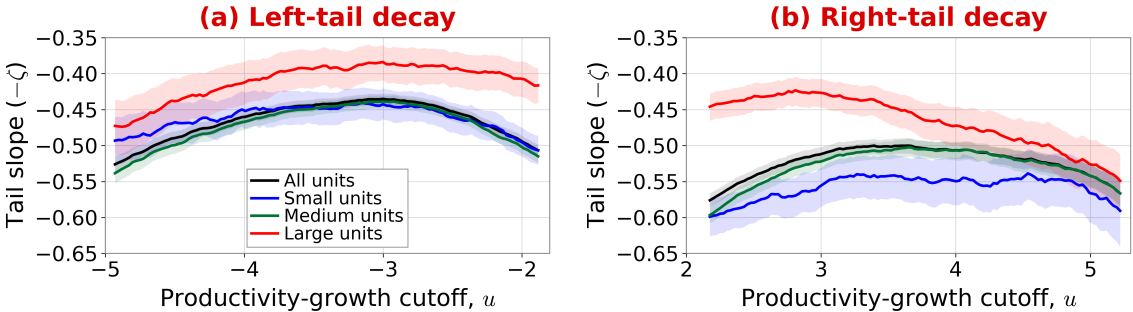
Closer to zero means slower decay and greater average distance beyond the cutoff.

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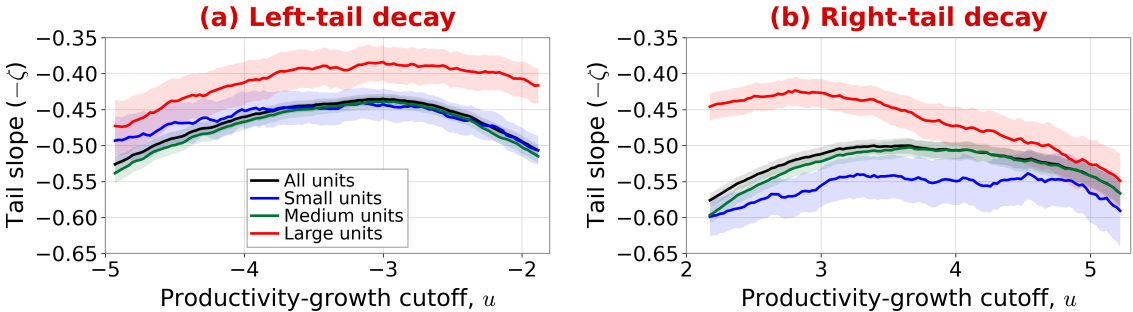
Farther out: move left in panel (a), right in panel (b). Shading: 95% pointwise plant-cluster bootstrap intervals.

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Farther out: move left in panel (a), right in panel (b). Shading: 95% pointwise plant-cluster bootstrap intervals.

1. **Nonconstant decay.** Outer tails steepen; a fixed-slope double Pareto misses this pattern.
2. **Large units reach farther:** lower conditional growth left, higher right. The gap is larger on the right.
3. **Similar total incidence** ($\sim$5.7%, group-specific 2 SD). Large units enter left less often, right more.

Statistical Model

A law of motion for productivity growth

A Latent-Regime Model of Productivity Growth

Each production unit can switch among $K = 3$ unobserved regimes.

Transition probability

$$\Pr(R_{i,t+1} = r \mid R_{it} = s, \mathcal{H}_{it}) = p_{sr}(g_{it}, d_{it})$$

g: productivity growth
d: log sales, standardized within year
$\mathcal{H}_{it}$: observed history through t

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Emission equation

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Normal within each regime.

Normal mixtures flexibly approximate conditional densities (Norets, 2010)

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Cubic growth, linear size, with interactions

$$\Phi(x, d) = (1, d)' \otimes (1, x, x^2, x^3)'$$

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Transitions: growth and size determine switching

$$p_{sr}(g, d) = \frac{\exp\{\eta_{sr}(z, d)\}}{\sum_{\ell=1}^{K} \exp\{\eta_{s\ell}(z, d)\}}, \quad \eta_{sr}(z, d) = \kappa_{sr}' \Phi(z, d), \quad \eta_{sK} = 0.$$

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$u(g)$ maps g's 1st–99th percentiles to $[0, 1]$ and clips outside.

146 free parameters ($K = 3$)

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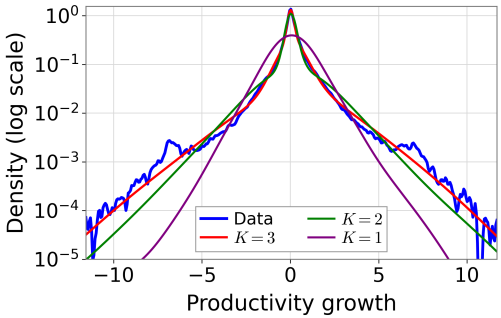
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Joint maximum likelihood. **Forward recursion** evaluates the likelihood.
Forward–backward smoothing recovers each unit's regime probabilities at each date.

Three Regimes Improve Distributional Fit

Density: core and tails



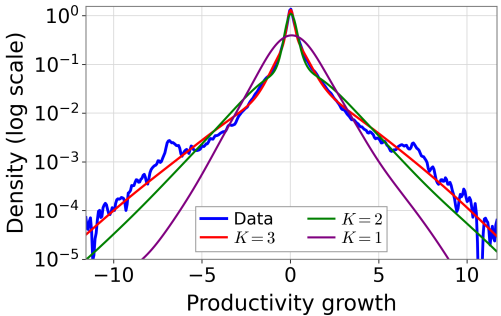
Density forecasts using $t-1$ information

$$\overline{f}_K(g) = N^{-1} \sum_{i,t} \widehat{f}_K(g \mid \mathcal{H}_{i,t-1}) .$$

Use lagged growth, size, and filtered history.

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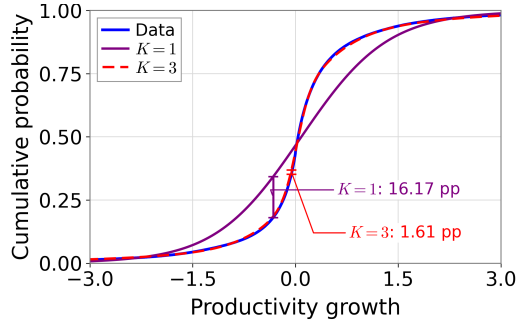


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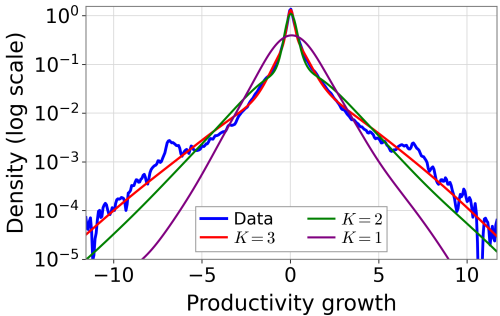
CDF: maximum discrepancy



Same N = 267,285 unit-year observations. Parameters estimated on the full panel. In-sample comparison.

Three Regimes Improve Distributional Fit

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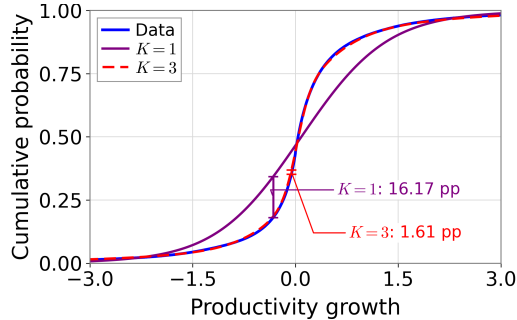


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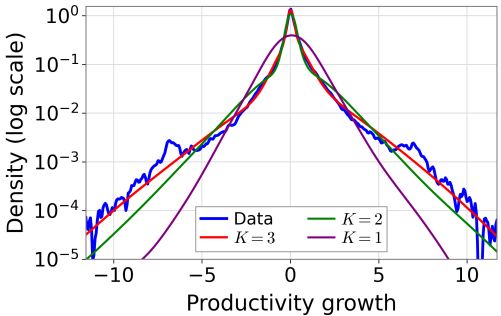
$$D_K = \sup_g \left| \widehat{F}_{\text{data}}(g) - \overline{F}_K(g) \right|$$

Regimes	$K=1$	$K=2$	$K=3$
CDF KS (pp)	16.17	3.22	**1.61**

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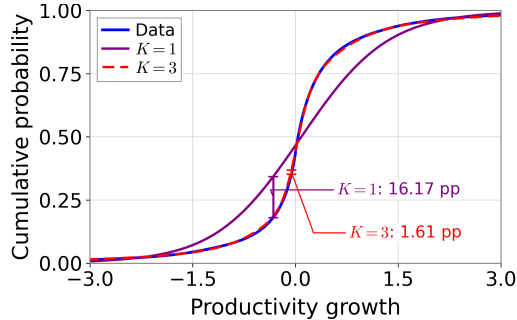


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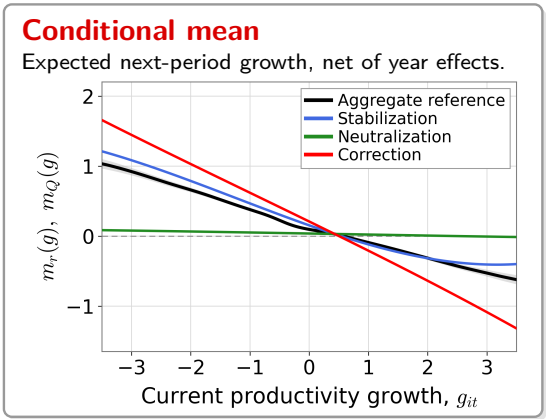
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CDF KS (pp)	16.17	3.22	**1.61**

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Three regimes capture the concentrated core and tail shape.

The maximum CDF gap falls by **90% versus $K=1$** and **50% versus $K=2$**.

Regime Names Reflect the Shape of Mean Reversion

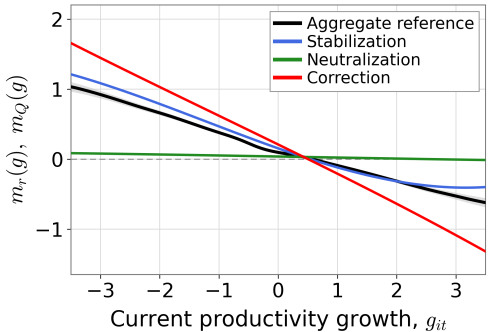


Regime curves fix size at each regime's probability-weighted mean and remove annual location effects. Black: aggregate reference m_Q. Gray: simultaneous confidence bands for the aggregate.

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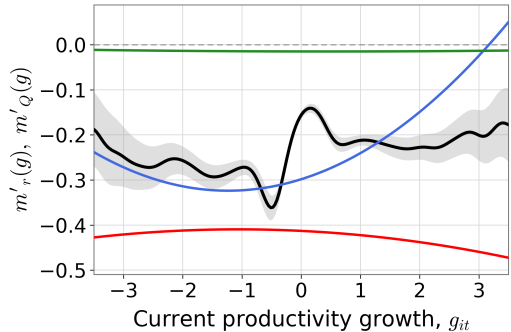
Conditional mean

Expected next-period growth, net of year effects.



Marginal slope

More negative slopes imply stronger reversal.

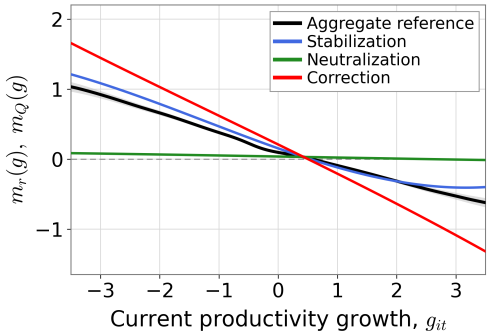


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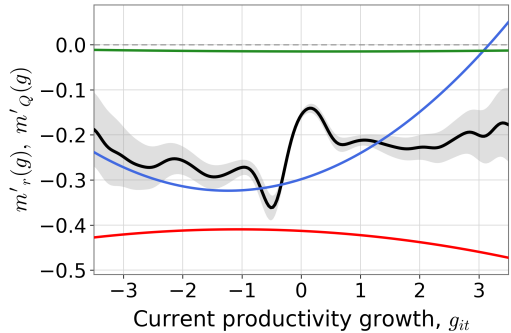
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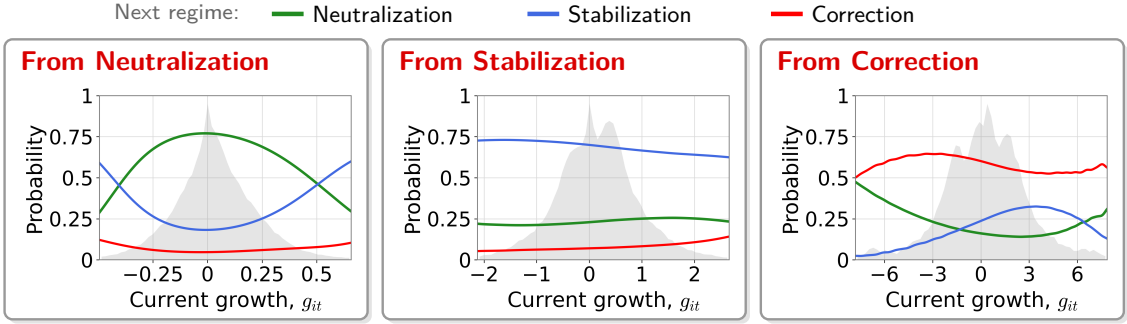
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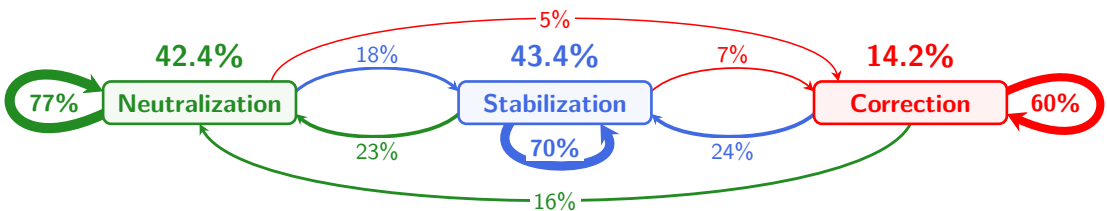
Stabilization: moderate, nonlinear reversal that weakens after large positive realizations.

Neutralization: next growth stays near zero, with little dependence on current growth.

Correction: a steep, nearly linear response producing the strongest reversal.

Transitions Across the Three Regimes





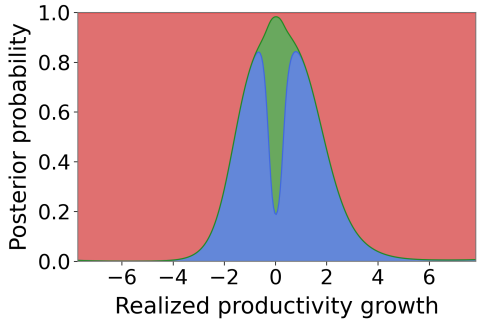
At neutralization's tails, the probability of moving to stabilization rises to about 60%.

Where the Regimes Concentrate

Stabilization Neutralization Correction

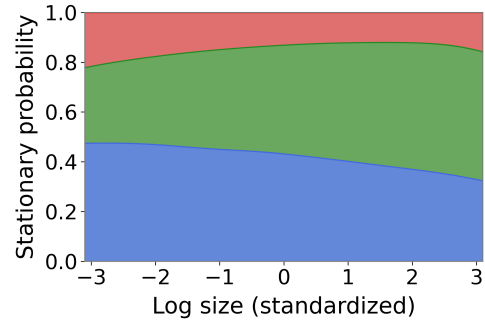
Across productivity growth

Posterior shares from the pooled density.



Across firm size

Stationary composition implied by local transitions.



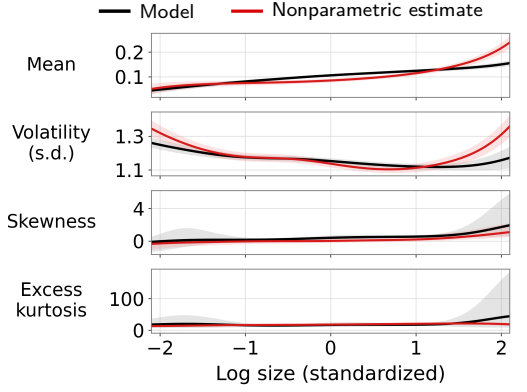
The vertical thickness of each colored band gives that regime's probability.

Growth: neutralization in the core, stabilization on the shoulders, correction in both tails.

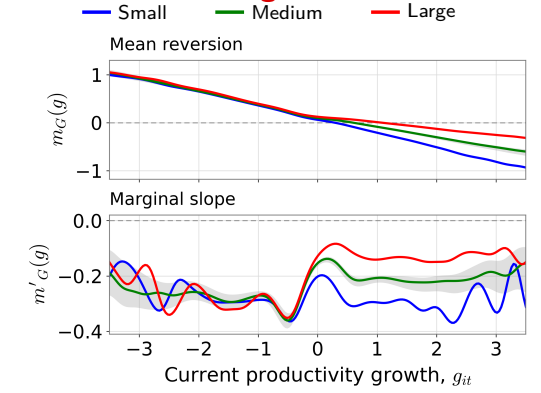
Size: more neutralization (30% to 52%) and less stabilization among larger firms.

Model Fit: Moments and Nonlinear Buffering

Moments across firm size



Daedalian buffering



Moments: the model captures the broad patterns of all four moments across firm size.
Buffering: stronger for large firms, with liftoff, corridor, and fallout nonlinearities.
Dynamic gap: the model overstates Icarian liftoff for small firms.

Aggregate Productivity Risk

From microeconomic shocks to aggregate outcomes

Microeconomic Shocks Track Aggregate Fluctuations

Cost-based Domar aggregation

$$G_t^x = \frac{100}{\rho_t^x} \sum_{i \in \mathcal{R}_t^x} \widetilde{\lambda}_{i,t-1} x_{it}, \qquad x \in \{g, \varepsilon\}$$

g: productivity growth; ε: innovation.
$\widetilde{\lambda}$: Domar weight; ρ: retained weight share.

Allocative-efficiency changes are omitted: a correlation of one is not expected.

Microeconomic Shocks Track Aggregate Fluctuations

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Correlation with aggregate data

Full Table 5

	Productivity	Innovation
KLEMS gross-output TFP	0.708	0.839
ASI real output per person	0.574	0.517
NSO real manufacturing GVA	**0.756**	**0.870**

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Full Table 5

	Productivity	Innovation
KLEMS gross-output TFP	0.708	0.839
ASI real output per person	0.574	0.517
NSO real manufacturing GVA	**0.756**	**0.870**

Co-movement with manufacturing GVA

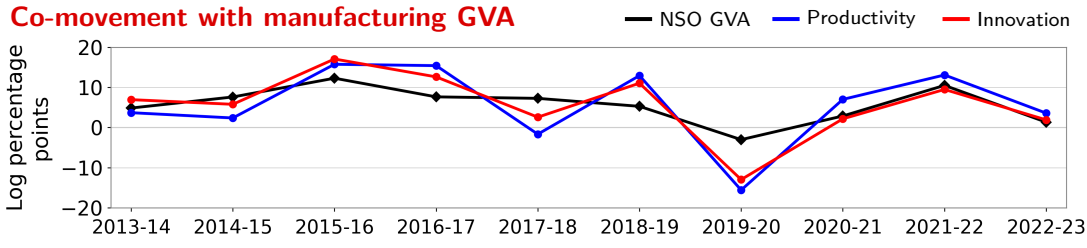
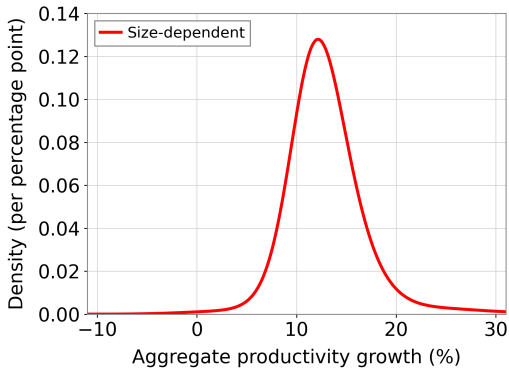


Figure 13 adds constants to align the aggregates' means with NSO GVA; these shifts leave correlations unchanged.

Nonnormality, Size, and Aggregate Risk

Cost-based Domar aggregation



Aggregate moments and tail risk

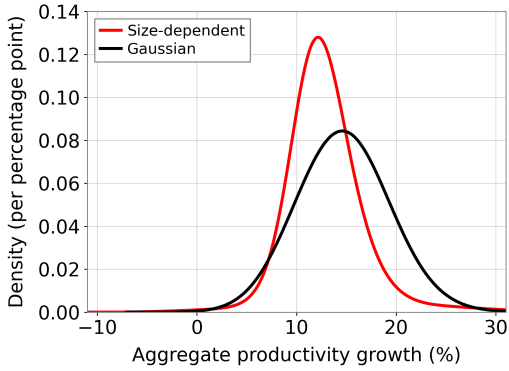
	Size-dependent 3 regimes
Mean (%)	**12.95**
SD (pp)	**4.25**
Skewness	**1.382**
Excess kurtosis	**6.917**
$\Pr(\|z\| > 2)$	**4.47%**
$\Pr(\|z\| > 3)$	**1.89%**

Separately estimated laws; same cost-based Domar weights; one billion simulated economies per model.

$X = 100[\exp(G) - 1]$.

Nonnormality, Size, and Aggregate Risk

Cost-based Domar aggregation



Aggregate moments and tail risk

	Size-dependent 3 regimes	Gaussian Keeps size 1 regime
Mean (%)	12.95	14.69 (Δ +1.73)
SD (pp)	4.25	4.73 (Δ +0.48)
Skewness	1.382	0.084 (Δ -1.298)
Excess kurtosis	6.917	0.021 (Δ -6.896)
$\Pr(\lvert z\rvert > 2)$	4.47%	4.54% (Δ +0.07)
$\Pr(\lvert z\rvert > 3)$	1.89%	0.29% (Δ -1.60)

Δ vs. red (unrounded); mean, SD, and tail changes in pp.

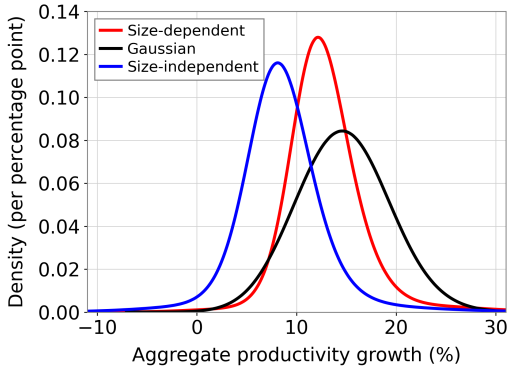
Separately estimated laws; same cost-based Domar weights; one billion simulated economies per model.

$$X = 100[\exp(G) - 1].$$

One Gaussian regime (keep size): SD rises 0.48 pp; three-SD tail probability falls 1.60 pp.

Nonnormality, Size, and Aggregate Risk

Cost-based Domar aggregation



Aggregate moments and tail risk

	Size-dependent 3 regimes	Gaussian Keeps size 1 regime	Size-independent 3 regimes
Mean (%)	12.95	14.69 (Δ +1.73)	8.53 (Δ -4.42)
SD (pp)	4.25	4.73 (Δ +0.48)	4.64 (Δ +0.39)
Skewness	1.382	0.084 (Δ -1.298)	0.676 (Δ -0.706)
Excess kurtosis	6.917	0.021 (Δ -6.896)	5.724 (Δ -1.193)
$\Pr(\lvert z\rvert > 2)$	4.47%	4.54% (Δ +0.07)	4.91% (Δ +0.44)
$\Pr(\lvert z\rvert > 3)$	1.89%	0.29% (Δ -1.60)	1.90% (Δ +0.01)

Δ vs. red (unrounded); mean, SD, and tail changes in pp.

Separately estimated laws; same cost-based Domar weights; one billion simulated economies per model.

$$X = 100[\exp(G) - 1].$$

One Gaussian regime (keep size): SD rises 0.48 pp; three-SD tail probability falls 1.60 pp.
Remove size dependence (keep three regimes): mean falls 4.42 pp; SD rises 0.39 pp.

Microeconomic Dynamics Shape Aggregate Tail Risk

Large units shape the aggregate distribution

Cost-based Domar weights transmit the distinctive productivity dynamics of large units.

Red versus size-independent: higher mean (12.95% vs. 8.53%) and excess kurtosis (6.917 vs. 5.724).

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Probabilities of Aggregate Technology Growth

Aggregate growth	Size-dependent	Gaussian	Size-independent
Below 0%	**0.3150%**	**0.0553%**	**2.4670%**
0% to 5%	1.1900%	1.7560%	14.3900%
20% to 25%	**2.7930%**	**11.3900%**	**1.1130%**
Above 25%	1.7450%	1.6580%	0.7760%

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Above 25%	1.7450%	1.6580%	0.7760%

Gaussian understates contraction risk and overstates 20–25% growth

Contractions: 0.315% under the red model vs. 0.0553% under Gaussian (**5.7 times as likely**).
Growth of 20–25%: Gaussian assigns 11.390% vs. 2.793% under red (**4.1 times as likely**).

Weights determine whose shocks matter. Their dynamics and shape determine aggregate risk.

Conclusion: Granularity Is Distributional

Physical productivity reveals non-Gaussian risk

Concentrated core, fat tails, and nonlinear productivity dynamics.

Firm size shapes recovery and persistence

Larger producers **recover more after losses** and **retain more after gains**.
A three-regime law captures these broad patterns.

Micro dynamics change aggregate risk

Relative to the full model, the conditionally Gaussian specification is overly optimistic:
It **understates contraction risk** and **overstates the probability of 20–25% growth**.

Large firms matter through both their economic weight and their distinctive productivity dynamics.

Appendix

Multi-product Plants

Allocate shared inputs using product gross-sales shares

Applies to: plants producing multiple 7-digit commodities.

1. Compute each product's share of plant gross sales.

$a_{bg} = \dfrac{S_{bg}^{\text{gross}}}{\sum_{h \in \mathcal{P}_b} S_{bh}^{\text{gross}}}$. b: plant; g: commodity; $\mathcal{P}_b$: products of plant b.

2. Allocate every shared input with the same share.

$Z_{bg} = a_{bg} Z_b$. Z: shared input quantities, expenditures, or capital values/costs.

3. Preserve the plant totals.

$\sum_{g \in \mathcal{P}_b} Z_{bg} = Z_b$. Product-level output quantities remain as reported.

Totals are preserved before subsequent outlier adjustments.

Back to mapping

Inventory Discrepancies

Reconcile unusually large differences between production and sales

Applies to: units whose production–sales quantity gap exceeds 30% of production.

1. Flag the discrepancy.

$$\frac{|y_i - \widetilde{y}_i|}{y_i} > 0.30. \qquad y_i\text{: quantity produced; } \widetilde{y}_i\text{: quantity sold.}$$

2. Reset sold quantity for flagged units.

$$\widetilde{y}_i^{\,*} = y_i, \qquad \Delta s_i^{*} = y_i - \widetilde{y}_i^{\,*} = 0. \quad \text{Other units remain unchanged.}$$

3. Recompute the unit price, keeping nominal sales fixed.

$$p_i^{*} = \frac{S_i}{\widetilde{y}_i^{\,*}}. \quad \text{Preserve reported sales } S_i \text{ and production } y_i.$$

2022–23: 2,653 of 83,500 production units affected (3.18%). A star denotes the adjusted value.

Back to mapping

Inverse-markup Outliers

Cap the cost-to-output ratio without changing relative cost shares

Applies to: units with $\mu_i = C_i/(p_i y_i) > 10$, after the inventory correction.

1. Compute a common scaling factor.

$a_i = \min\left\{1, \dfrac{10}{\mu_i}\right\}$. Units with $\mu_i \leq 10$ have $a_i = 1$.

2. Scale all expenditure components proportionally.

$C_{ij}^* = a_i C_{ij}$. j covers intermediate inputs, primary factors, and capital costs.

3. Recompute the ratio; keep the cost composition.

$\mu_i^* = \min\{\mu_i, 10\}$, $\dfrac{C_{ij}^*}{C_i^*} = \dfrac{C_{ij}}{C_i}$. Quantities remain unchanged at this step.

2022–23: 296 of 83,500 production units affected (0.35%).

Back to mapping

Commodity Prices and Elasticities

Recover substitution elasticities and normalize common prices

Applies to: each commodity–year, using inventory-adjusted quantities and capped costs. $S_g = \sum_i S_i$ and $\sigma_i = S_i / S_g$ within commodity g.

1. Recover the elasticity from output, sales, and costs.

$$\eta_g = \frac{\sum_{i \in \mathcal{N}_g} p_i y_i / S_g}{1 - \sum_{i \in \mathcal{N}_g} \sigma_i \mu_i}.$$

Inventories enter through produced versus sold quantities.

2. Aggregate sold quantities with the CES elasticity.

$$Y_g^0 = \mathrm{CES}_{\eta_g}\left(\{\widetilde{y}_i, \sigma_i\}_{i \in \mathcal{N}_g}\right), \qquad p_g^0 = \frac{S_g}{Y_g^0}.$$

3. Set the reference price from input expenditure and quantity.

$$p_g = \frac{\sum_j E_{jg}^x}{\sum_j x_{jg}}, \qquad Y_g = \frac{S_g}{p_g}, \qquad \frac{\sum_j x_{jg} \tau_{jg}^x}{\sum_j x_{jg}} = 1.$$

Denominator floor: 0.001; $\eta_g \geq 1.001$; Cobb–Douglas limit near one.

Back to mapping

Input Unit-value Outliers

Bound extreme input wedges while preserving expenditure

Applies to: intermediate inputs (τ^x) and primary inputs (τ^ℓ): labor, other domestic primary inputs, and direct imports. Capital is separate.

1. Compare the input unit value with its common reference price.

$\tau = \dfrac{E}{pq}$. E: expenditure; q: input quantity; p: common reference price.

2. Bound the wedge between 0.1 and 10.

$\tau^* = \min\{10, \max\{0.1, \tau\}\}$. Wedges inside the interval remain unchanged.

3. Adjust quantity to preserve the observed expenditure.

$q^* = q\dfrac{\tau}{\tau^*}$, $E = \tau pq = \tau^* pq^*$. Keep E and p fixed at this step.

Expenditure on adjusted observations (2022–23): intermediates 0.051%; primary factors 0.251%.

Back to mapping

Appendix: Correlation with Aggregate Data

Table 5: all aggregate benchmarks

Back to main slide

Aggregate measure	Productivity shock	Innovation shock
KLEMS gross-output TFP	0.708	0.839
ASI real output per person	0.574	0.517
KLEMS real gross output	0.446	0.463
ASI real output	0.449	0.447
NSO real manufacturing GVA	**0.756**	**0.870**
KLEMS VA-TFP	0.738	0.854
KLEMS real GVA	0.697	0.827
ASI real output per worker-day	0.508	0.380
Manufacturing IIP	0.291	0.403

Construction and interpretation

Pearson correlations over ten years, FY2013–14 to FY2022–23. Innovations: five-fold plant-held-out AR(2).

Trim the lowest and highest $\lfloor .075 n_t \rfloor$ observations separately by shock and year. Keep retained shocks uncentered and divide each weighted sum by the retained share of matched Domar weight.

$$\rho_t^x = \frac{\text{Retained Domar weight}}{\text{Matched Domar weight}}, \qquad x \in \{g, \varepsilon\}$$

$\mathcal{M}_t$: matched; $\mathcal{R}_t^x$: retained.
$\widetilde{\lambda}_{i,t-1}$: beginning-of-period Domar weight.

Descriptive validation: allocative-efficiency changes are omitted, so perfect correlation is not expected. Input growth and coverage also differ across aggregate benchmarks.

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